

Traffic Participants in the loop: A Mixed Reality-Based Interaction Testbed for the Verification and Validation of Autonomous Vehicles

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Abstract—In order to verify and validate autonomous vehicles, testbeds integrating the whole system from perception to actuation are necessary. Above all, this applies to the assessment of the autonomous vehicle's performance in the presence of vulnerable road users' behavior: While experiments for the interaction between autonomous vehicles and pedestrians in critical traffic scenarios are hardly conductable in reality, virtual experiments often suffer from plausibility. In order to solve this issue, we present a mixed reality testbed for the verification and validation of autonomous vehicles faced with realistic road user behavior in critical, worst case traffic scenarios. We achieve this by registering an immersed pedestrian and the automated driving function within a common environment model, providing challenging traffic scenarios. The testbed is applicable within different integration levels of the automated driving function and enables a high level of behavioral realism. The testbed is evaluated qualitatively and discussed within a concrete use case.

I. INTRODUCTION

During the last decades, robotics research made great progress towards intelligent and *Autonomous Mobile Systems* (AMS), also enforcing progress in the field of *Autonomous Vehicles* (AV). Typical *Advanced Driver Assistance Systems* (ADAS) more and more support the driver in his tasks, e.g. with *Automated Emergency Braking* (AEB) or *valet parking* assistance. The continuous integration of these assistance systems enables vehicles to handle more and more traffic situations autonomously, see [1], [2], [3]. Nevertheless, they are only exposed to a minimal subset of traffic situations. So, the verification and validation (V & V) of these systems under varying traffic participants' behavior remains an unsolved problem.

For the (V & V) of an autonomous vehicle, its highly automated driving function, hereinafter called *system under test* (SuT), has to be exposed to a wide range of diverse traffic scenarios. Above all, the need for exposure holds for road users with a critical behavior in traffic situations with shared spaces, such as ordinary urban junctions with pedestrians, bicyclists and vehicles as well as parking facilities. There, road users are characterized by a wide range of actions and

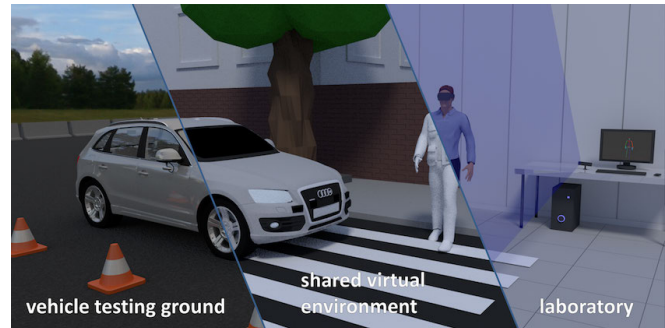


Fig. 1. Mixed reality testbed with a common environment model shared between spatially separated road users. The left part of the image represents the real world environment of the vehicle captured by on-board sensors. On the right side, a human with virtual reality goggles is captured by a motion tracking system in a laboratory. The overlap represents the environment model, where the automated vehicle and the pedestrian safely interact.

interactions, especially by non-compliant behavior. In particular, pedestrians are of interest: They move more slowly compared to other traffic participants, but are more likely to suddenly change their intention and direction.

An autonomous vehicle must take this ability into account. Obviously, *collision mitigation* is one of the skills when the system has to avoid any collision with a vulnerable pedestrian. Moreover, the *prediction of non-compliant behavior* is another system functionality. The system has to model, predict and understand any behavior of a pedestrian, regardless whether it is intrinsically or extrinsically motivated. Finally, it has to be capable of *pedestrian to vehicle cooperation* in unregulated traffic situations, such as negotiating one's right of way. Therefore, this has to be testable safely. Furthermore, many algorithms for autonomous vehicles rely on machine learning to cope with the variety of traffic scenarios. From the point of view of the principle of inductive learning, there is a great need for data-acquisition testbeds, enabling the generation of datasets of realistic and critical traffic scenarios.

In order to tackle this issue, we propose a mixed reality testbed, that focuses on behavioral-realistic human-machine interaction in critical traffic scenarios. This is achieved by the generalization of the *X-in-the-loop* framework and a human immersed into the role of the pedestrian, and finally, the introduction of a joint environment model, where spatially separated road users are faced together in a critical traffic

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scenario. The variation of this joint environment enables the generation of challenging and critical scenarios by for example increasing traffic flow or varying occlusions. In summary this facilitates the exposure of the SuT to the random, unforeseeable behavior of an immersed person, whereas, the safety of the traffic participants is assured at any time.

This paper is structured as follows: Sec. II deals with an overview of the related research. Then in Sec. III the design and architecture of the testbed is presented, followed by the presentation of its realization in Sec. IV. We present concrete applications and discuss the interaction testbed in Sec. V. Finally we close the paper with a summary and an outlook for future research questions in Sec. VI.

II. RELATED WORK

The scientific scope of the present paper is located in the research field of verification and validation of intelligent robots and autonomous vehicles. Since especially for autonomous mobile systems testbeds are applied, we also consider mixed reality testbeds for robotic applications. We first describe the *Milgram's Continuum* [4], a concept where all the following approaches can be categorized into. It shapes a taxonomy for the fluent transition of real and virtual worlds to provide novel environments where physical and virtual objects co-exist, see Fig. 2. It can be applied e.g. for collaboration of distributed user groups [5] or interaction with simulated groups [6]. In the field of robotics, mixed reality environments are used to provide testbeds for robots. In [7] a general framework is presented, which enables interactions between mobile robots and obstacles at different scales. It is demonstrated, how a real mobile robot is able to react to a virtual obstacle and vice versa. In [8] a mixed reality framework for the development of *unmanned aerial vehicle* is presented.

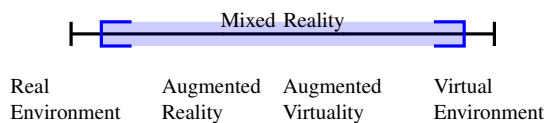


Fig. 2. Milgram's Continuum [4] ranges from the reality over different mixed reality instances to virtual environments.

Different strategies exist in order to assess the performance of an autonomous vehicles' capabilities. Real tests, for example, are classified as beyond of the left side of the continuum: On proving grounds test rigs are used in order to reenact critical vehicle-to-vehicle or vehicle-to-pedestrian interactions by pulling pedestrian dummies using portal rigs or cable pedestrian rigs over the autonomous vehicle's track [9]. The approach is restricted to generate traffic scenarios with simple interactions and it is neither applicable for complex traffic situations nor for flexible cross-site application nor for fast, structural modifications of the scene. On the other hand, public road tests or field operational tests [1], [2] are performed to validate the functionality under the stochastic nature of experienced traffic scenarios. They are

time-consuming, expensive and risky for human and material and detected misbehavior is mostly not repeatable. Moreover, the underlying ground truth situation is hardly observable.

Beyond of the right side of the continuum, virtual environments are applied for autonomous vehicle testing: Different commercial frameworks, such as IPG CarMaker [10] or PreScan [11] as well as Open Source frameworks, like CARLA [12] exist that enable integrating the automated driving function into a virtual environment. The system's performance is assessed under observation of the feedback of the vehicle mechanics as well as the environment which is an adequate approach especially in the beginning of the V-Model, applied in the automotive development process.

Clearly, the simulation models' quality also affects the experiment results, as discussed in [13]. So the performance assessment of an automated driving function significantly depends on the task suitability of the road users' behavior model as well as of its correct parametrization. Our framework avoids this issue by replacing model-based behavior, such as provided by frameworks like Menge [14] with a real immersed pedestrian in a data-driven manner.

Typical V & V concepts are categorized as logical combinations and partitionings of real and virtual components [13], referred to as *X-in-the-loop*: Within *hardware-in-the-loop* (HiL) testing the algorithm is integrated in an electronic control unit (ECU) as a real world component and is stimulated by signals from a simulated environment and fed back into a subsequent simulated component (e.g. powertrain). Within *driver-in-the-loop* tests a virtual driver is replaced by a human driver within a static or dynamic driving simulator to gain insight about the interaction between a potential driver and the automated driving function. In *vehicle-hardware-in-the-loop* (VEHIL) a real, physical vehicle with the sensor setup is fixed on a chassis dynamometer, and a robotic moving base is controlled as a target vehicle [15]. Finally, *vehicle-in-the-loop* (ViL) [16], [17] combines the former approaches: An experimental vehicle equipped with sensors, actuators and the highly automated driving function is moved on a free proving ground. The SuT is stimulated by simulation and reacts to virtual obstacles, for instance. The human as a potential driver or test engineer is integrated in the testing process for instance using robust headtracking [18] for augmented and virtual reality based immersion of the human [19]. In [20] the idea to include real pedestrians into the autonomous vehicle testing process has been sketched, but lack of a concrete realization and evaluation issues.

Beside the aforementioned *X-in-the-loop* instantiations, also real sensor data, such as from KITTI [21] or virtual sensor data such as from the *Virtual KITTI* [22] or SYNTHIA dataset [23] are applicable to stimulate a driving function in an open-loop manner. Stimulating the system on a sensor-close level is desirable since it enables the assessment from perception to actuation. In *vehicle-in-the-loop* testbeds, mixed reality generated sensor data is applicable in closed-loop testing, as we have demonstrated for stimulating camera-based [24] and LiDAR-based [25] components.

III. ARCHITECTURE OF THE INTERACTION TESTBED

A. Generalization of *X-in-the-loop*

In simulation frameworks the behavior of traffic participants is typically mapped to either trigger-based replay of fixed trajectories, e. g. when falling below a certain distance to the SuT, or model-based behavior. However they can be applied to challenge the vehicle, but lack either responsiveness in the sense of realistic feedback or critical behavior is hardly configurable. Also, realistic, multi-layered behavior, such as a pedestrian that is about to cross the road and waits for a deceleration of the approaching vehicle to make sure that his intention has been understood, is hardly transferable to simulation within a general model.

From a testing point of view, the assumptions on which the behavior modeling is based may not correspond to the real world. Many models assume an ideal perception of other agent's position and behavior, for instance. For these reasons, we replace the necessity of the validity assumption of behavior models by data-driven immersion to validate and verify the SuT in the presence of complex and unforeseeable interactions between the autonomous vehicle and an immersed human.

Hence, our approach is not only consistent, but rather conclusive with the aforementioned *X-in-the-loop* approach, see Fig. 3. We explicitly generalize the testing of a highly automated driving function as a SuT and additionally including the interacting road user for observation and evaluation into the testing loop. Consequently, our approach is applicable along the entire V-Model-based development process in the automotive domain, where components are developed and tested iteratively as models, software and as hardware components, finally being tested as ECU on testbeds or in the vehicle: In early phases the SuT is stimulated as a software component and later stimulated in a rapid prototyping vehicle by the immersed human as a vulnerable road user.

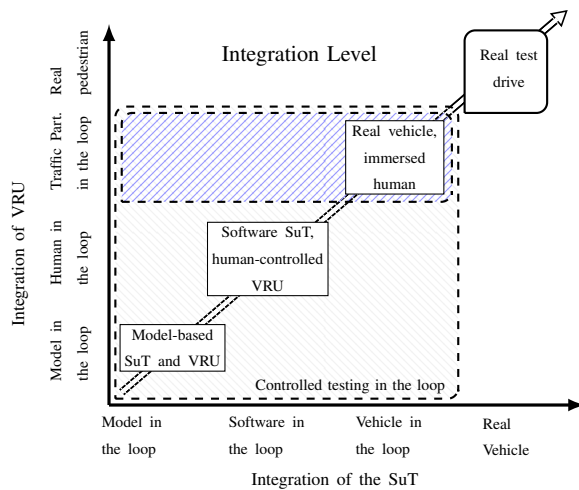


Fig. 3. Generalization of *X-in-the-loop*: The gray shaded region indicates, where virtual testbeds typically are applied. Our framework opens up a new subspace close to the real test drive in order to evaluate driving functions in the presence of an immersed human (blue shaded space). The arrow indicates an increasing degree of realism and significance of evaluation.

B. Generalized Avatar Model

Therefore, the generic concept of an avatar model is introduced: It seamlessly replaces a model-based behavior of a traffic participant in the simulated environment by a concrete instance of a real world entity and its real behavior, see Fig. 4. This holds for the replacement of a virtual vehicle by its real counterpart as well as for the replacement of the virtual pedestrian by the immersed human.

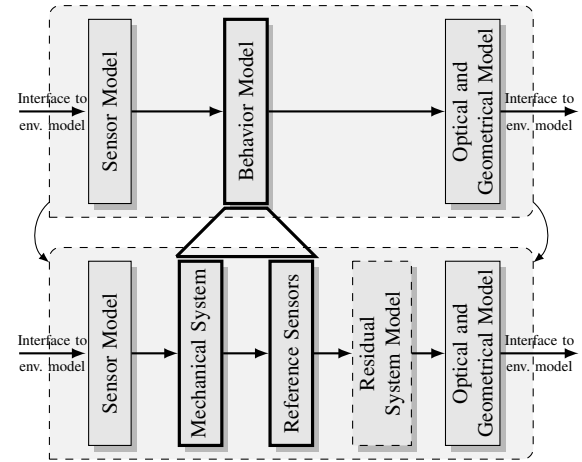


Fig. 4. Concept of the generalized avatar model (below), which is applied to replace behavior models seamlessly with real entities.

Our avatar model is characterized by mainly three components:

- **Sensor Model:** It reflects how the current environment model is discretized and transformed in order to be perceived by the model itself or the real entity.
- **Reference Sensors:** Instead of an explicit behavior model, an entity's behavior is derived from the real entity by observing its spatio-temporal behavior. Additional quantities that are relevant to the interaction are observed.
- **Optical and Geometrical Model:** A virtual entity represents optical and geometrical properties of the avatar.

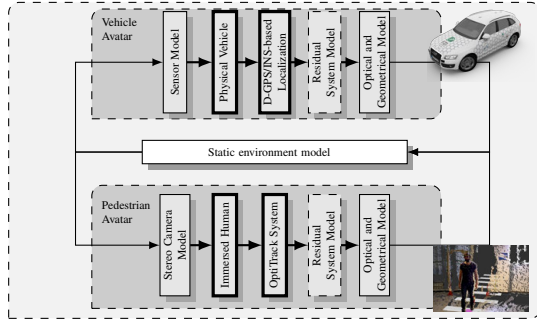
Nevertheless, a fourth component may be necessary: The residual system model maps residual dependencies between observable and non-observable quantities, for which an appropriate reference measurement system doesn't exist. If we cannot observe the human joints' trajectories or the vehicle's wheel twists, we may derive them from the observed vehicle velocity within an additional model, whose output is handled in the optical and geometrical model.

Therefore, two basic assumptions have to be made: Firstly, the reference sensors are precise enough with respect to their spatial and temporal resolution and a sufficient precise mapping to the avatar quantities is available. Then the *ground truth property* is fulfilled. Secondly, the sensor models are precise enough in order to stimulate the immersed human as well as the SuT appropriately. If these properties are not fulfilled, the testbed itself propagates potential errors to the system's performance assessment.

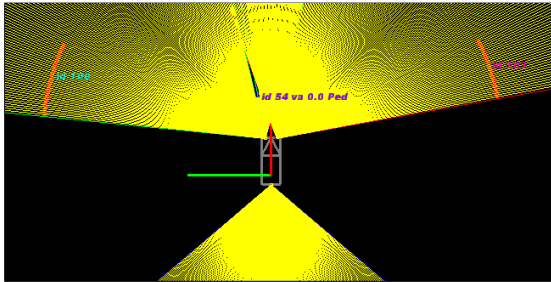
C. Integration of the system under test

In early steps of the development process our testbed eases the testing of interactions and training of an autonomous vehicle in the presence of pedestrians. Hereby, the SuT is integrated as *model* or *software* coupled with a vehicle dynamics model and stimulated by a sensor model. In our framework, we consider physically motivated sensor models such as a *Light Detection And Ranging* (LiDAR) sensor model based on raycasting as well as high level sensor models, such as obstacle or pedestrian's joints list.

In order to validate the SuT in a rapid prototyping system, it is integrated on a real physical vehicle in the loop, see Fig. 5(a). Controlled by the automated driving function, it receives sensory input from the environment model. Therefore, the SuT is stimulated by sensor signals, whereby real sensor signals are either fully replaced by virtual signals or augmented in the signal's feature space, see Fig. 5(b). Its spatio-temporal behavior in terms of position and orientation, velocity and acceleration is measured by a localization system, such as a D-GPS/INS system.



(a) The spatially separated traffic participants, both the autonomous vehicle with the SuT and the pedestrian are tracked by reference sensors. Then, their behavior is mapped to their respective avatar models in the environment model.



(b) Stimulation of the SuT by mixed reality LiDAR sensor signals: The real LiDAR signature is augmented with a signature of a virtual pedestrian. For a detailed description see [25].

Fig. 5. Instance of our testbed with a vehicle-in-the-loop instantiation.

Embedding the algorithms in the real, physical vehicle provides several advantages: Firstly, the SuT is running very closely to the final vehicle platform. Secondly, no assumptions about the vehicle are made and thus replaced by the real vehicle's dynamics. Finally, the vehicle's movement can be stimulated by external impacts, such as wet surfaces resulting in loss of friction, increasing the level of realism as much as possible.

D. Integration of pedestrians

Beyond usual ways of controlling a pedestrian model using keyboard or game controllers in multi-user driving simulators by *human-in-the-loop*, see Fig. 3, we directly focus on a real, immersed person's behavior to provide unforeseeable, random and model-free behavior. In terms of our general avatar model, see Fig. 4, an appropriate sensor model, stimulating the immersed person as well as an appropriate reference measurement system, observing the person's behavior, are needed:

For the immersion of a human, a suitable stereo camera model, considering a baseline between both eyes and a shifted view frustum, is needed in order to create a three dimensional impression. The virtual camera then visualizes the avatar's virtual perspective rendered from the simulated environment model. Such camera models are used for common virtual reality goggles, such as Oculus Rift or HTC Vive. They are applicable to immerse the human, but limit the usage to the available cable length. Therefore, our approach is based on wireless display systems, such as smartphones, which enable to be immersed over a possibly large space by motion tracking systems.

The SuT dictates the level of inclusion it has to be exposed to by our framework: The necessary level is provided by either a direct observation with the reference sensors or by deducing them within a residual system model. If a component is integrated, which estimates whether a pedestrian will cross the street or not by observing its gait, our framework provides this data by direct observation of the pedestrian's joint angles or by deriving the joint angles with a motion synthesis as a residual system model, see Fig. 6.

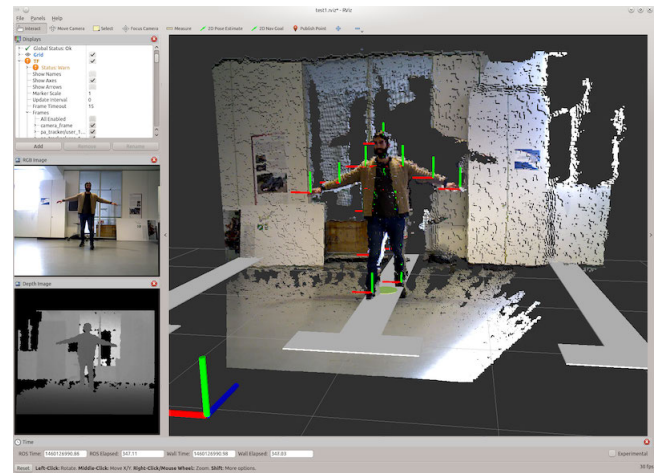
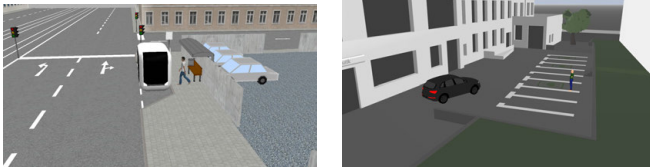


Fig. 6. Early version of the framework: Tracking a human with an RGB-D camera limits its application to a small working space and frontal orientations, whereas skeleton tracking enables observing the joint movements. The white bars indicate virtual parking space markings of the scenario Fig. 7(b).

E. Simulated environment model

The environment model aligns the spatially separated real pedestrian and the physical or virtual autonomous vehicle to a common reference frame, where risky interactions between

both are performed safely without any risk. It defines the relevant scenario elements, such as the fundamental road structure with roads, crossings, linked regulatory elements such as traffic lights or signs, buildings and static obstacles. Also additional dynamic traffic participants may be included in terms of traffic flow. In general, the elements depend on the type of scenario, such as highway, rural or urban sceneries. Scenarios can be set up twofold, in a fictitious or in an accurate manner, see Fig. 7. In order to create worst case scenarios for both the immersed person and the vehicle with the SuT, in particular, we include and vary semi-static obstacles manually, such as parking vehicles to cause restricted visibility for the immersed human as well as the vehicle's sensors.



(a) Modeled fictitious scenario with a public transportation stop. (b) Parking scenario modeled according to a real parking site.

Fig. 7. Urban scenarios characterized by potential vehicle to pedestrian interactions.

In the instance of our testbed with the higher integration level, both workspaces of the physical vehicle and pedestrian have to be registered: This means, that the immersed vehicle and pedestrian shall not exceed their respective dedicated spaces while moving and interacting in the joint environment model. The maximum spatial extension of the scenario is determined by the spatial extent of the free proving ground and the human tracking space in the laboratory.

F. Communication and Analysis

For the application of *software- to hardware-in-the-loop* testing, the system is built up on a co-simulation platform, running on a single or on multiple computers, being synchronized over a wired network. For *vehicle-in-the-loop* testing, a distributed co-simulation and wireless communication is necessary to bridge the different, spatially separated working spaces of the vehicle and the pedestrian, see Fig. 5(a). Therefore, only the reference measurements, such as the current poses of the traffic participants are exchanged, whereas the residual models and sensor stimuli and renderings are computed locally near the laboratory, where the pedestrian is tracked, and on a separate ECU in the experimental vehicle.

As the communication medium in our testbed architecture, we propose to use a IEEE802.11p/ITS-G5 based V2X communication system. For exchanging position and velocities, the ETSI-standardized *Cooperative Awareness Messages* (CAM) can be used between the respective vehicle's and the pedestrian's sensor simulation. Since the standard defines the frequencies for the transmission of CAMs, the V2X ECUs have to operated outside of the specified transmission frequency for sending CAMs or additional, traffic participant-

specific extrapolation techniques have to be applied in order to increase the measurement updates on each side.

Synchronization and thereby traceability of the chronological sequence of the observation quantities then is guaranteed by stamping the different quantities with GPS time instead of an artificial simulation time. This enables the replay of scenarios for open-loop testing of the system's components, further system's performance assessment, road users' behavior analysis, but also the use in machine learning, which relies on synchronized, multivariate data.

IV. REALIZATION

Our realization is based on the modular robot simulation *Gazebo* [26] and the middleware *Robot Operating System* (ROS) [27], since both are flexible regarding the integration of models as well as entire SuTs due to quasi-standardized interfaces for sensor data and actuator commands. We have realized our concept using model plugins. So, the embedded participants can be controlled by model-based behavior or updated with the measured poses by the reference systems.

A SuT is instantiated within an URDF-based [28] vehicle model with different automated driving functions. In case of the data-driven mode, the localization of the avatar model is derived from the D-GPS/INS system and imprinted to the avatar. The sensor data then is provided to our SuT. The pedestrian is realized programmatically in the same way: The human is located by an Optitrack Tracking System, consisting of a camera array with six Flex13 cameras in total, enabling an ellipsoid workspace of around 4×2 meters, with a specified subcentimeter precision of 0.5mm. Wireless virtual reality glasses has been built up using a smartphone and trackable markers, see Fig. 8. The virtual pose of the head is located, imprinted to the virtual pedestrian's head to render a stereo view from the virtual environment model, which is compressed and transmitted to the smartphone.

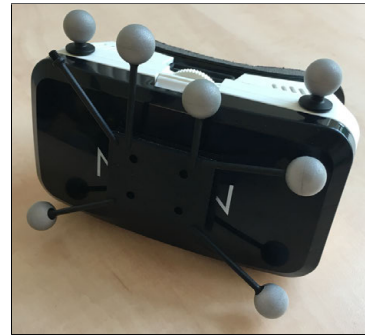


Fig. 8. Realization of smartphone-based virtual reality glasses, being tracked with the attached Optitrack markers.

For the application in *vehicle-in-the-loop* testing traffic participant dedicated *Gazebo* instances are connected. The communication is handled via a *road side unit* (RSU) placed near the tracking area and a V2X ECU in the vehicle itself. Both are configurable to send CAMs outside the specified frequency. Both are configurable to send and receive CAMs. In order to reduce latency, the sending frequencies of the RSU and the ECU are increased to surpass the specification.

V. EVALUATION

Since it is challenging to evaluate the quality of a testbed, which actually is designed for a system's performance assessment itself, we take the approach to evaluate it by a qualitative evaluation as well as a concrete use case to demonstrate the benefits of our concept.

A. Qualitative Evaluation

In order to enable a qualitative evaluation of testing methods, we gathered the criteria *observability*, *safety*, *throughput*, *flexibility*, and *realism* and compared three testing methods introduced in Sec. II to our novel testbed. The results are visualized in the Kiviat diagram, Fig. 9:

Observability defines the testing system's ability to provide insight about ground truth quantities of all entities, involved in the test: It is obvious, that in a complete virtual environment, we have full observability of all simulated entities, whereas on real test drives, only vehicle related quantities are observable. So we state, that the mixed reality approach provides a similar degree of observability compared to dummies on proving grounds.

Safety defines the risk of damage or even injury imposed on the SuT and additional pedestrians: The strict spatial separation inherent to a mixed reality approach allows the reduction to a minimum. The safety factor can be seen as an inverse measurement of inducible risk. Our framework therefore enables maximum worst case testing with almost no damage of any kind, similar to the evaluation in a virtual environment. It is obvious that safety cannot be assured for real test drives, whereas on proving grounds damage is limited to the vehicle and the pedestrian dummy used.

Throughput defines the number of experiments that can be performed given a fixed time interval. As our approach is strictly limited to real-time operation due to a human immersion, pure virtual validation can achieve much higher throughput. Nevertheless, interactions between the pedestrian and the SuT can be performed quite iterative depending on the given problem. Since we have to reset the vehicle's as well as the pedestrian's state to initialize the next scenario. It is clear that throughput and especially coverage in scenario-based testing is hard to determine while driving on the real road.

Flexibility defines the effort required to create novel and unforeseen traffic situations, the SuT is exposed to. As well as within virtual validation, the mixed reality environment can be modified effortlessly to vary situations from urban scenery to accidents on highway scenarios, or to include obstacles and occlusions in order to worsen the scenario. Due to the spatial separation, even reckless behavior may be provoked without impact on safety. Also, fictitious as well as known critical static scenarios can be built up. For real test drives appropriate route planning increases the flexibility to face the vehicle with different scenarios. The flexibility on proving grounds is limited to the capability of the proving ground, e. g. its road structure.

Realism refers to the level of behavioral realism with respect to the scenario context the SuT is stimulated with.

Our proposed method allows for a gradually adaptation of the realism to meet the requirements of the SuT. Since the evaluation results of virtual validation depends on the quality of the integrated models, especially the behavioral models, our approach faces the vehicle with realistic behavior of pedestrians. Since behavioral models are used for pulling pedestrian dummies over the track, our approach provides a higher degree of realism, which is solely outperformed by challenging pedestrian behavior faced with the real vehicle.

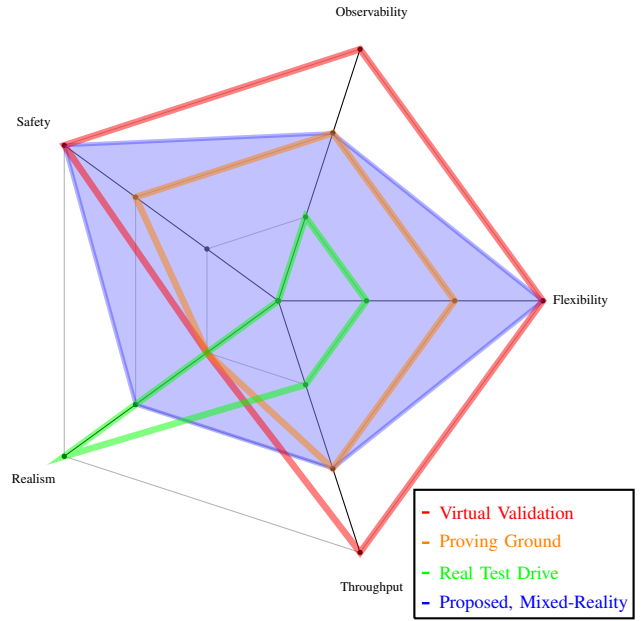


Fig. 9. Qualitative analysis based on five dimensions: observability, safety, throughput, flexibility and degree of behavioral realism.

We argue that this qualitative analysis reveals that our approach provides a flexible, general and realistic testbed for generating critical scenarios with vulnerable road users. Our framework provides similar properties as tests on proving grounds with pedestrian dummies but is more flexible. For the immersed human's perception as well as the vehicle's sensors a mediator with the means of photo-realistic image generation or other sensor models is still necessary to obtain realistic data from this common environment, since only the behavioral models of the traffic participants are replaced by real world entities.

B. Use Case: Parametrizing an Adaptive Cruise Control

In order to verify our concept, we demonstrate the application of our testbed to validate an exemplary driving function in an urban scenario in a very early step of the development process. Therefore, we have mapped a real urban district in terms of the road layout, modeled with the use of Lanelets [29] and static buildings derived automatically from OpenStreetMap [30] and textured manually to provide a credible and plausible scenario. As an exemplary SuT, a LiDAR-based adaptive cruise control (ACC) and collision mitigation system based on the *Intelligent Driver Model* (IDM) [31] is applied. It is characterized by a minimal safety distance

s_0 from bumper to bumper, a minimal time distance T_0 , a desired target velocity v_0 , achievable with a maximum acceleration a_0 and braking deceleration b_0 . Additionally, the model considers an acceleration component δ , usually set to 4.0. The vehicle's new acceleration a is calculated such as:

$$a = \frac{dv}{dt} = a_0 \left(1 - \left(\frac{v}{v_0} \right)^\delta - \left(\frac{s^*(v, \Delta v)}{s} \right)^2 \right) \quad (1)$$

$$\text{with } s^*(v, \Delta v) = s_0 + v \cdot T + \frac{v \cdot \Delta v}{2 \cdot \sqrt{a_0 \cdot b_0}} \quad (2)$$

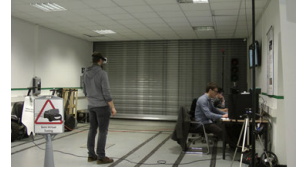
The immersed person was told to cross the street in front of a parking bus to reach the tram, see Fig. 10. Stepping in and out of the sensor scope, the immersed human was able to challenge the SuT very easily, so that crashes could be reproduced without the danger for the human of being harmed. This approach enabled us to optimize the IDM's model parameters, especially the safety distance and the maximum deceleration. Although no strict repeatability is given, our concept enabled us to experience the situation from a pedestrian's perspective in a safe way to evaluate, what vehicle behavior is perceived to be pleasant for the person interacting with it.

VI. CONCLUSION

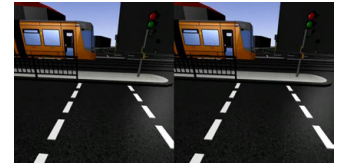
We conclude that our testbed enables the evaluation of a highly automated driving function for verification and validation within different integration levels in worst case scenarios in the presence of realistic behavior of spatially separated road users. We want to emphasize, that especially for validation purposes, when a SuT is evaluated if it fulfills its intended application, our testbed is able to face a SuT with realistic interactions of vulnerable road users. This holds especially for the performance assessment of collision mitigation and avoidance functionality the prediction functionality of compliant and non-compliant behavior and, finally, functionality for *vehicle to pedestrian cooperation* for negotiating unsignalized traffic settings, for example.

The strict separation of the simulated environment model and the real system guarantees safety, even when the real physical vehicle is involved. For instance, having the real vehicle and other road users safely separated, allows the human participants of the test to act in ways that are irresponsible or even dangerous under normal conditions such as ignoring traffic lights or taking the autonomous vehicle's right for passage. That way, testing and data-acquisition can be obtained in form of a *serious game*.

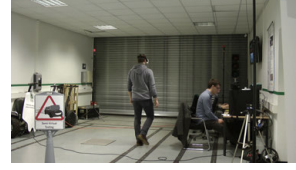
This work has revealed several research issues, we want to pursue with our future work. Since, we made the assumption that by the use of reference systems our realization is a *ground truth* generating system and that the sensor models are precise enough to stimulate both, pedestrians and the SuT, we focused on the architecture and realization of the interaction testbed. So the question remains how measurement errors of the poses of the vehicle and pedestrian are propagated through the system into the evaluation results. Therefore, a detailed examination is necessary to assess the reference property of every element in the processing chain,



(a) The human is instantiated at the beginning of the scenario.



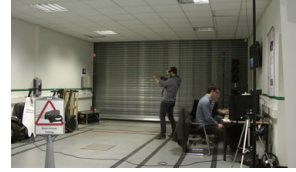
(b) The vehicle is not yet visible in the human's field of view.



(c) The human is told to cross the street in front of the approaching vehicle.



(d) The virtual vehicle is now visible street in front of the approaching in the virtual field of view. vehicle.



(e) Finally, the human steps in front of the virtual vehicle.



(f) The pedestrian recognizes how close the vehicle actually is.



(g) The misparametrized system comes to a halt too close to the pedestrian.

Fig. 10. An exemplary use case of our testbed in an early development step of an LiDAR-based longitudinal adaptive cruise control: The real human is immersed in a mapped environment of a real urban district.

such as the sensors but also the communication and taking into account the necessary precision of the automated driving function and the scenario itself. Also, we want to evaluate how detailed a sensor model for an immersed human has to be designed according to visual accuracy and photo-realism to make the immersed person forget about his or her immersion. These research question are currently pursued. Also, we start the second phase in order to include as many road users into the integration test, possibly transforming it into a large-scale test environment to let them interact in more complex urban scenarios, which enable to generate complex pedestrian group behavior on complex urban scenarios.

Despite the fact that the proposed framework has been described in the context of integrative testing of autonomous driving functions and data-acquisition of behavioral pattern data in critical traffic scenarios, it is not limited to this field only. Its principle and its applied technologies can be transferred to any field where collaboration with an autonomous system is considered hazardous or simply is impossible or inconvenient to realize.

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